



EE712 - Embedded Systems Design

Electrical Engineering IIT Bombay

Laboratory Experiment - III

Problem set: 3.2

January 30, 2025

Reaction Timer

Problem Statement:

- The program starts with a message on LCD as **Press switch SW1** on the first line and as **LED glows** on the second line. Turn on a LED.
- The user presses the switch. The LED is turned OFF.
- The program is expected to measure time between the instant the LED starts glowing and the instant the switch press is identified.
- The display on the LCD shows Attempt no # on the first line and the time in # ms on the second line. This message is displayed for 30 seconds.
- The program then goes back to step 1 again and repeats for five user attempts.
- After five attempts, display Now press SW2 on the first line and for average response time on the second.
- Note that **we need to account for 10ms delay for switch debouncing.**

Components to be used:

- TIVA C Board
- 16 X 2 LCD Display

General Instructions

- Initialize the LCD. Find detailed procedure in the attached pdf (Tips for Interfacing LCD). And also refer Peripheral Driver Library Guide for System functions (Chapter 27) and Timer functions (Chapter 30).
- Set up the Timer in continuous(periodic) mode. Load in a value so that it takes 1 millisecond. Make sure you enable the timer via SysCtlPeriphEnable() and also the Timer interrupt. Do NOT run the timer yet. Functions: TimerConfigure(); TimerLoadSet(); IntEnable(); TimerIntEnable();
- Change the corresponding Timer interrupt handler in tm4c123gh6pm startup ccs.c to your own function (say void Timer2IntHandler()). Write the prototype of this function with the other prototypes in tm4c123gh6pm startup ccs.c. Detailed instructions in next few slides.
- This function should count (using a global variable (say times)) the number of times it is called.

Instructions for Interrupt Usage:

- Open file tm4c123gh6pm startup ccs.c

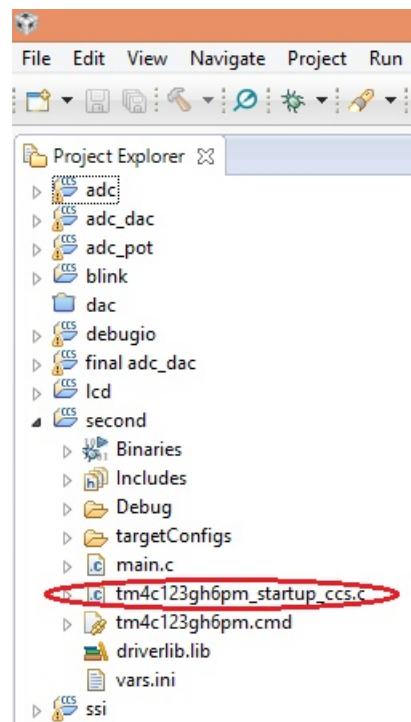


Figure 1: Screenshot 1

- In file tm4c123gh6pm startup ccs.c add declaration of your own interrupt function as static void. (We have added static void Timer2IntHandler(void);)

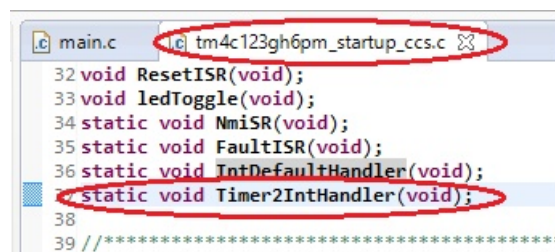


Figure 2: Screenshot 2

- In file tm4c123gh6pm startup ccs.c change the handler corresponding to the Timer to your own function (We have changed Timer2IntHandler)
- Define the function along with the other definitions.

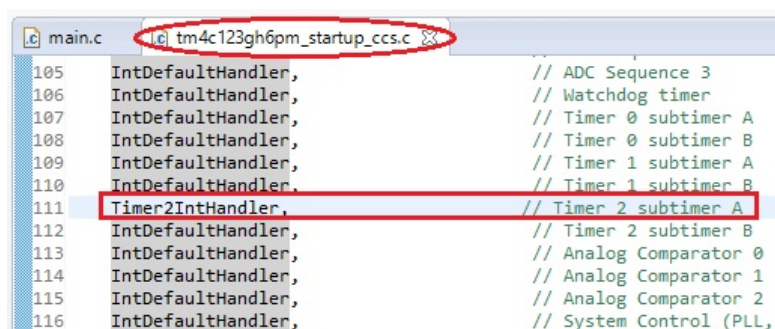


Figure 3: Screenshot 3

General Instructions (continued):

- Display MSG1. Wait for switch press. Glow GREEN LED. Display MSG2. Wait for sometime.
- Glow RED LED. Immediately run the timer. Wait in a loop until SW2 is pressed.
- As soon as SW2 is pressed, stop the timer. The value in the global variable times is the response time in milliseconds.
- Display response time.
- Repeat similar procedure 5 times. Finally calculate and display the mean response time.

Precautions and Tips:

- Timing is vitally important for LCD. Refer the manual and give appropriate delays.
- Unlock the switch as you did in previous labs.
- Make functions to separate the digits and convert to char. Bitwise OR of a digit(0 to 9) with 0x30 gives ASCII value of that digit.
- Use while() loops to wait for events.
- Do NOT call the interrupt function in the program. It is called directly when Timer reaches zero.
- The return type and arguments for interrupt function should be void.
- Calculate Timer load value considering the proper System Clock frequency. System Clock frequency may be obtained by SysCtlClockGet().
- Remember that SysCtlDelay(x) delays 3x clock cycles.
- Make sure you have included "driverlib/interrupt.h" and "driverlib/timer.h".
- In your interrupt handling function make sure you clear the interrupt before leaving the function, else it will be called immediately over and over again. Use the TimerIntClear() function.

References:

1. TM4C123GH6PM : TM4C123GH6PM_Datasheet
2. TIVA Driver Library: TIVA_Driver_Library
3. TM4C123G LaunchPad Workshop Workbook TM4C123G_LaunchPad_Workshop_Workbook